Building a Full Desktop in Go!

(windows, desktops, compositing, ...AI?)

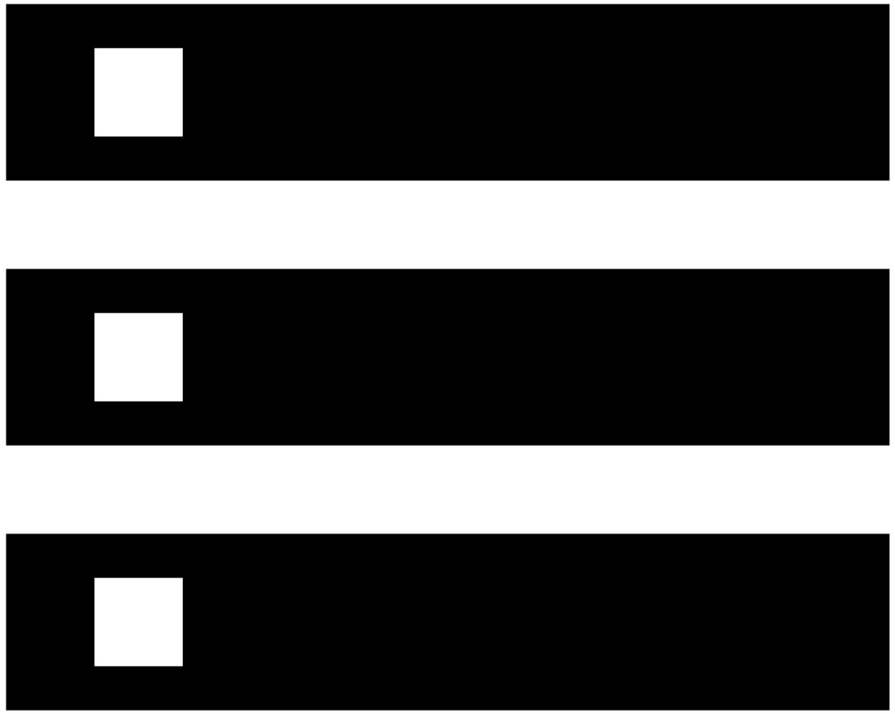
Another Desktop? Why?

- Linux on the Desktop!
- Performance AND productivity
- A developer oriented desktop
- The ultimate proof of **Fyne.io**



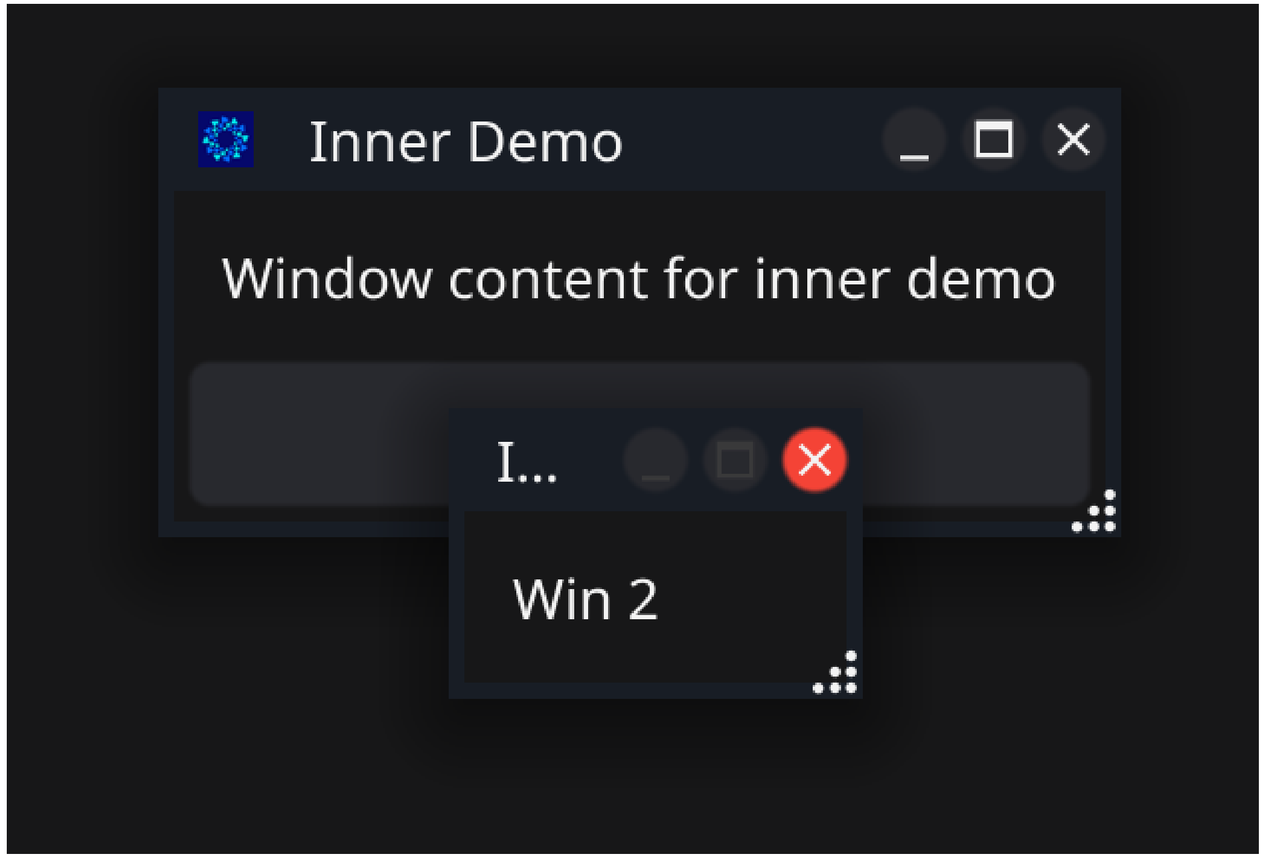
Architecture

- Server (X11 / Wayland)
- Window Manager
- Compositor
- Virtual Desktops
- App icons and launcher
- Modules & overlays



Window Manager

- Capture content / events
- Draw borders
- Stacking & position
- Iconify / Maximize
- Fullscreen / borderless



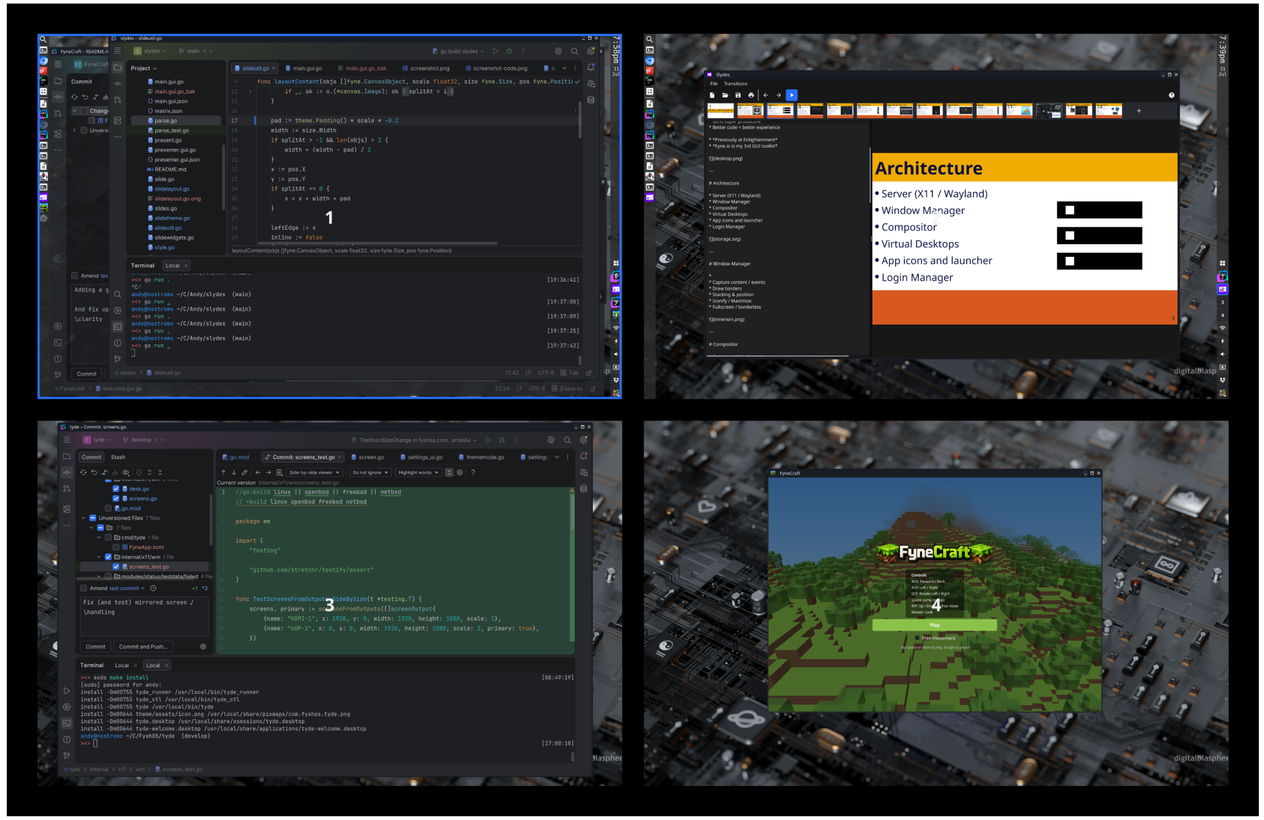
Compositor

- Capture visible windows
- Draw:
  - Background
  - Underlays (files, widgets)
  - Windows
  - Panels & overlays



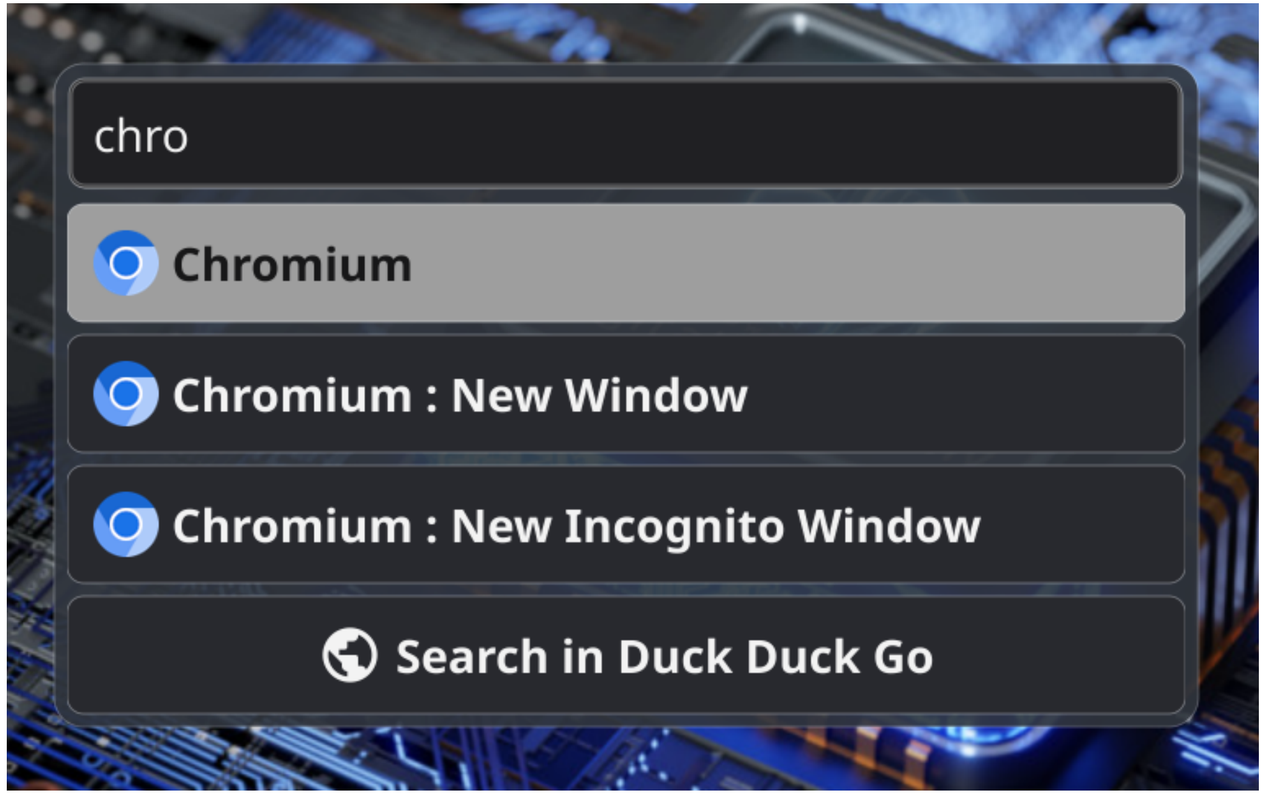
Virtual Desktops

- Just a big virtual space
- Move the image, not window
- Desktop pager, flip & zoom
- Key bindings and menus
- *And sync actual X11 windows*



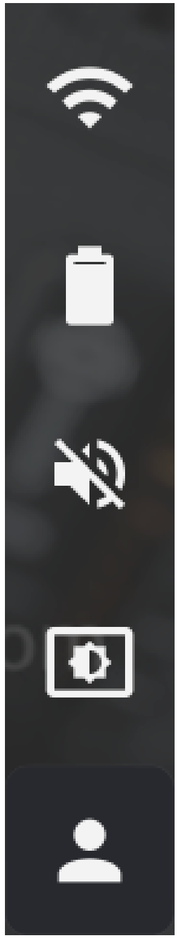
App icons and launcher

- Manage installed apps
- Load metadata and icons
- Launch apps and:
  - Conversions, calculations
  - Search the web
  - AI suggestions/chat



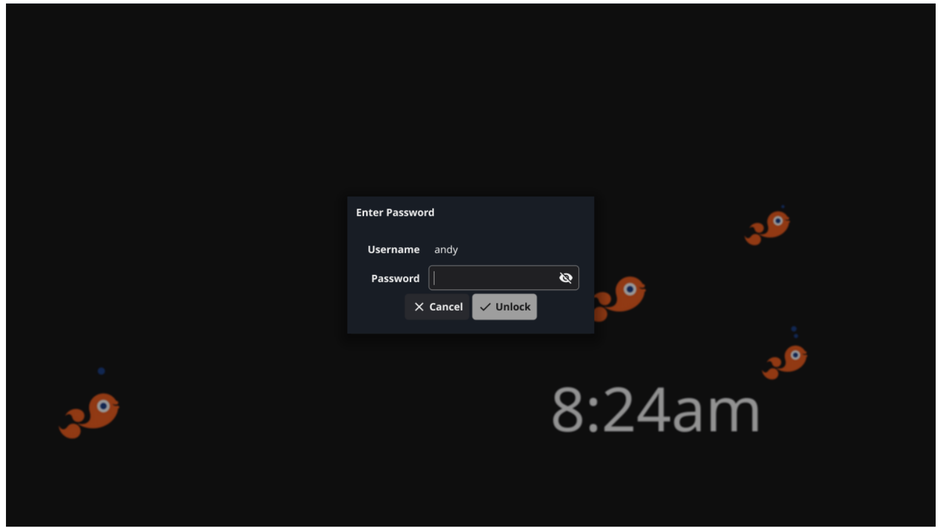
Modules

- Main extensibility point:
  - Status area (system tray, wifi etc)
  - Screen area (Fyles, backgrounds)
  - Launcher (Conversion, web search, AI)
  - Keyboard (brightness, airplane etc)
  - Overlays (flames, eSheep & more)



ScreenSaver

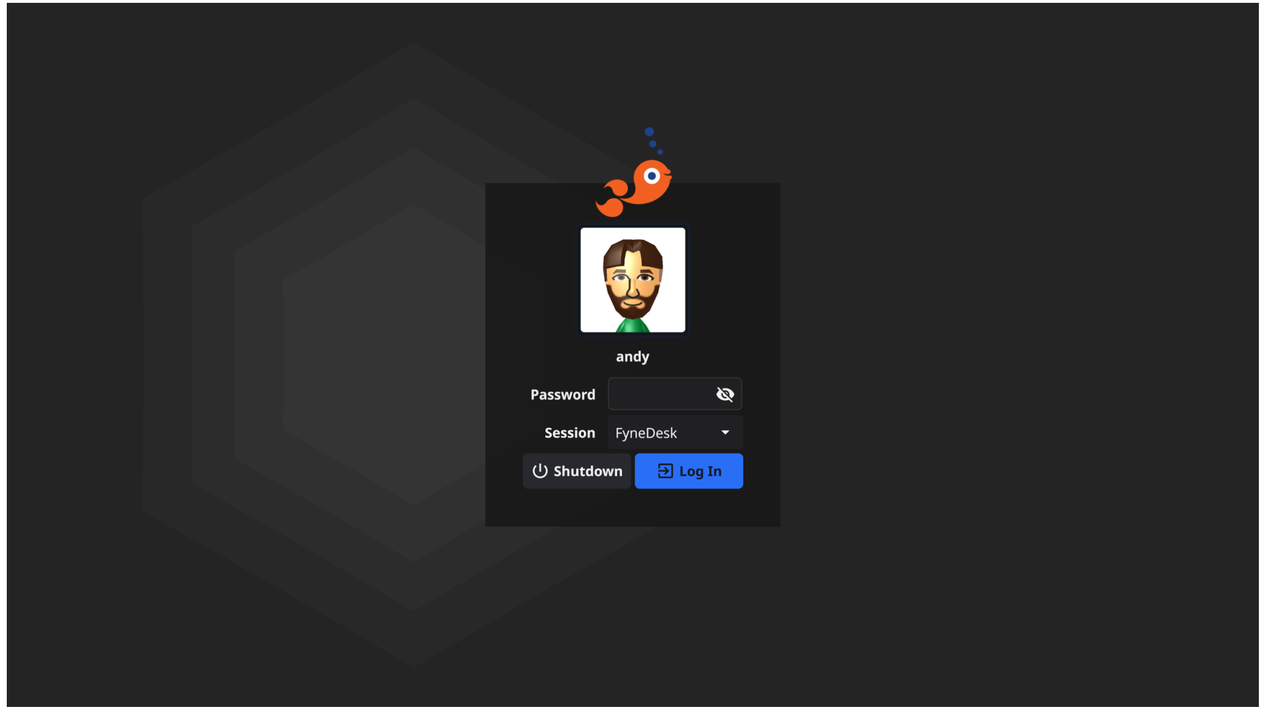
- Just another Fyne app
- One window per screen
- Fullscreen - swallow all events
- Password and fingerprint support
- Drop-in custom visual





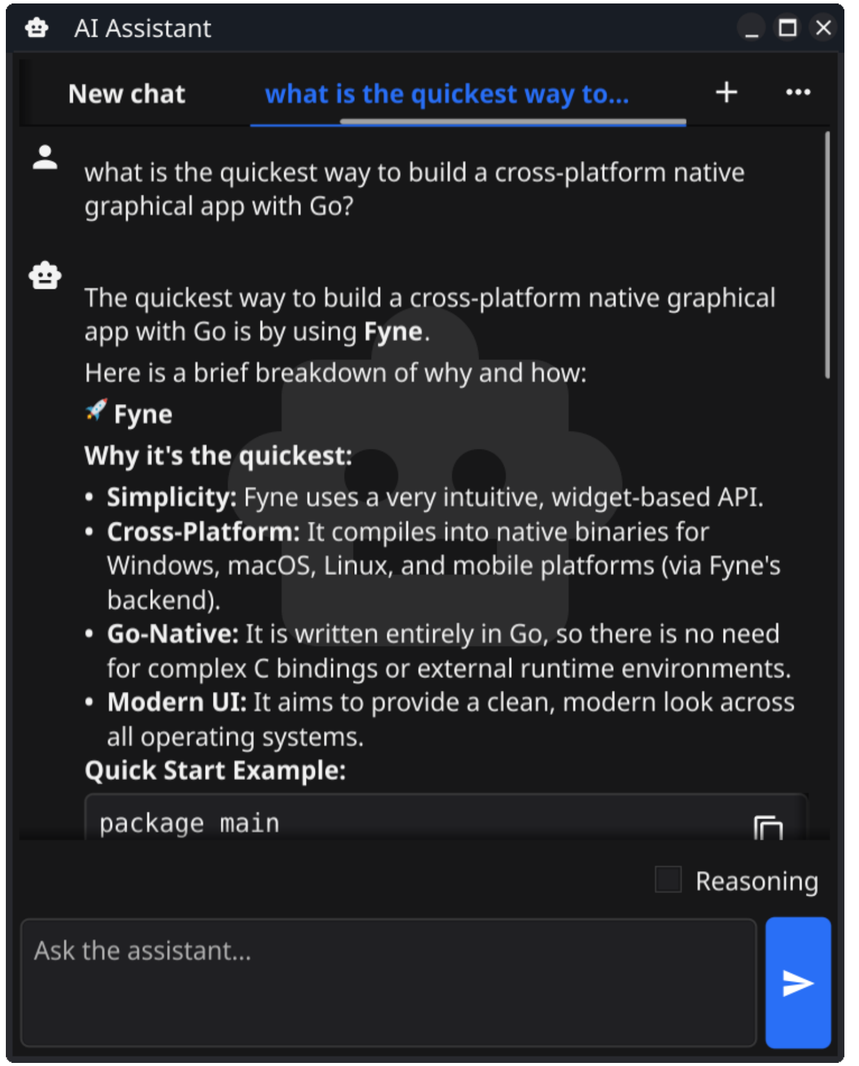
Login Manager

- App that launches desktop
- Handle users & logins
- Selecting desktop
- Reflect user configuration



AI Integration - local and frontier

- Local with Kronk or Ollama
- Connect OpenAI or Anthropic
- Fathom: Chat and quick answers
- Choose model and reasoning
- Control the desktop (in progress)

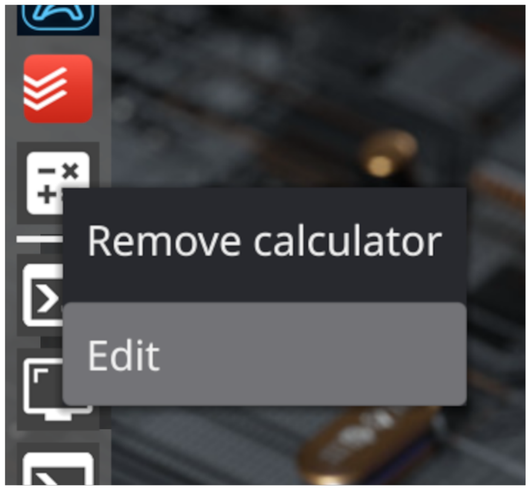


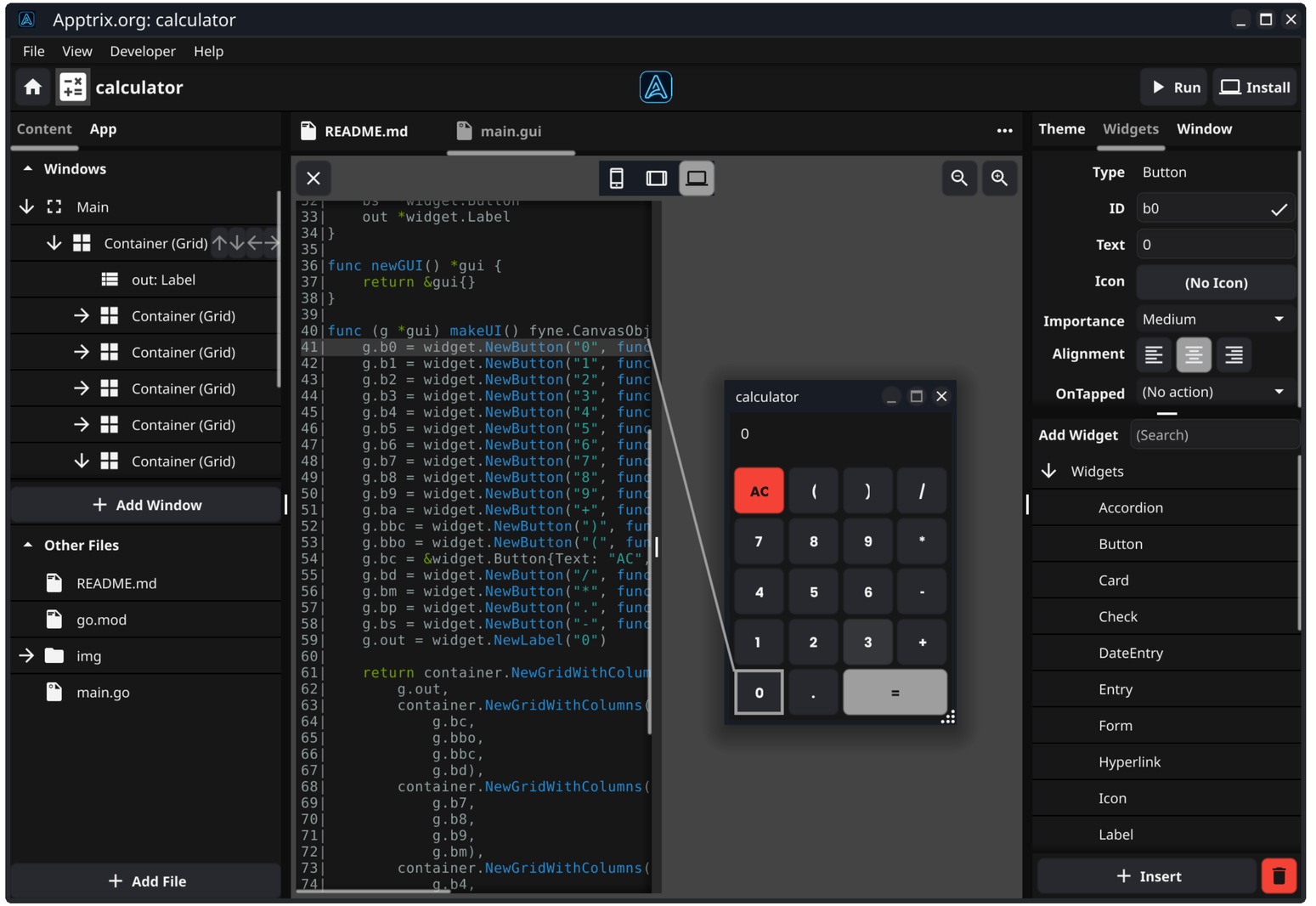
All Together = FyshOS

- tyde & fin
- backgrounds / appie
- dryvers / screens
- fancyfs / fyles
- fyshsaver
- ferry



Bonus: WYSIWYG App Editing





Find out more

- https://fyshos.com
- github.com/FyshOS/tyde
- #fyne
- @andydotxyz
- https://andy.xyz
- https://apptrix.ai

